From end states to pathways

Estimating acceptance, carrying it in agents, and finding robust road-pricing paths to a low-carbon Norway in 2050

No one has yet estimated road-pricing acceptance on real data, carried it in agents through the years, and let it decide which passenger and freight pathways survive. That is the gap this thesis sets out to fill.

Discrete choice (DCM) → agents with memory (ABM) → robust pathways under deep uncertainty (MORDM).

The work has started. The pipeline already runs end to end: a public repository, 89 tests, the Swissmetro choice model reproduced, two real road networks benchmarked. This is a foundation, not a finished thesis. Much is still to estimate and improve. But the groundwork is built, and it is open to check.
github.com/Astheno-sphere/computational_logistics

01 · THE PROBLEM

What an equilibrium cannot carry

Norway appraises transport with a national model that solves one equilibrium for one year. Ask it what 2050 looks like under a set of assumptions, and it answers with care. Ask it how the country gets there, and which routes fail on the way, and it has nothing to say. It was not built for that. TØI's own review agrees: the current models are static and macroscopic [toi1819]. The path to 2050 turns on what an equilibrium cannot hold. A car fleet that turns over slowly. Public acceptance that lasts or fades. Debt that comes due on a schedule.

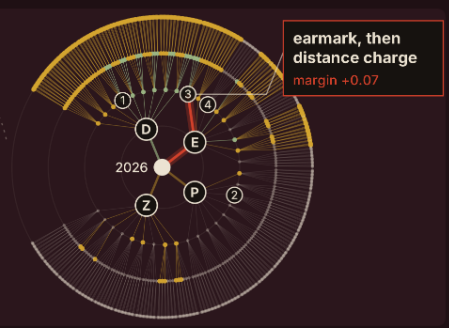
The zero-emission transition makes this concrete, and fiscal. Road pricing is one of the levers named for the next National Transport Plan, where the first TRANSPLAN report says the plan must be realistic, not a wish list [toi2172]. Yet the national reform has stalled. Tolls raised NOK 16.8 billion in 2025, about 62 percent of it in city packages, against toll-company debt near NOK 78 billion [svv25]. As cars go electric, the toll base erodes under the very policy that rewards them. And every revision meets a public that has voted against tolls before. In Bergen in 2019 a single-issue anti-toll party won 16.7 percent [snl19]. Whether a 2050 plan can work turns on three things: emissions, the public finances it rests on, and who it asks to pay. What decides it is the order the instruments arrive in, and whether people accept each step.

THE SPACE OF PATHS

A STRATEGIC MODEL
one point: 2050
the end state, if every assumption holds

THIS PROJECT
904 paths, each tested
leaf colour: futures on target, of 1,000

SAME 2050, DIFFERENT FATE
① distance, then package 67%
② package, then distance 36%
③ earmark, distance at 50% 63%
④ earmark, distance at 80% 23%



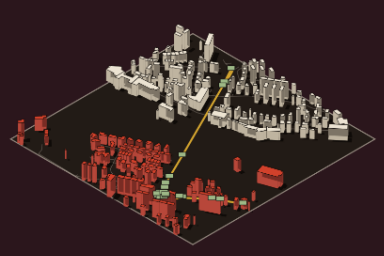
Five instruments can be switched on in many orders. A reduced set alone gives 904. Most fail on emissions, money, or acceptance. **The search on page 5 finds the few that hold.**

> *Main question. Which sequences of road-pricing instruments reach a good 2050 on emissions, finance and access while staying acceptable to the people and firms who pay, across deep uncertainty; and which signals tell planners when to switch?*
>
> **RQ1 · METHOD (A1)** Can a choice model built from real survey data drive agents that remember, so one 2050 end-state becomes a year-by-year path?
>
> **RQ2 · BEHAVIOUR (A2)** Does acceptance last once a charge is permanent, and can information move it on its own?
>
> **RQ3 · PLANNING (A3)** Which pathways, for passengers and for freight, stay robust; what acceptance margin does each survive; and which signposts warn before one fails?

A first measured result. At the Kristiansund tariff a zero-emission car with a tag pays NOK 19.04 instead of NOK 27.20. Each electric crossing earns NOK 8.16 less, a cut of 30 percent [vegamot]. Four in ten cars are already electric, so a crossing today earns about 12 percent less than it would in a fossil fleet. In an all-electric fleet it earns 30 percent less, and the debt does not move. This erosion returns on page 5 as the finance signpost. And the thesis finishes even if no survey or microdata ever arrives: the spine is my own data and published national ranges, as the ladder on page 5 shows.

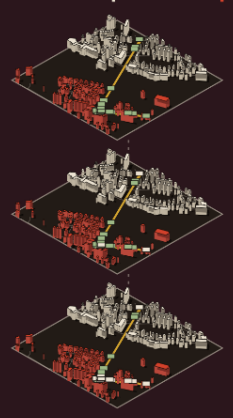
ONE ANSWER, OR A PATH

LEFT · THE STRATEGIC MODEL
One answer: 2050



one settled state, every car already where it ends up
no year in between · heights 3x

RIGHT · THIS PROJECT
The same place as a path



years from the prototype run, pathway E, page 5 · heights 3x

The left panel is what an equilibrium gives you: one settled 2050, every car already where it ends up. The right is the same town lived forward. The car turns electric. The toll bill falls. Acceptance rises, then dips. One branch survives, one breaks. **The gap between the two panels is the whole thesis.**

WHAT A CROSSING EARNS

> what one crossing earns, by fleet

standard car	NOK 34.00
with tag	NOK 27.20
zero-emission + tag	**NOK 19.04**
each electric crossing earns	**NOK 8.16 less, −30%**
today, at 40% electric	−12% vs all-fossil
all-electric fleet	−30%, debt unchanged
the base erodes, the debt holds	**~NOK 78 bn owed**

REAL · Vegamot tariff [vegamot]

The discount that rewards clean cars is also the leak in the fund that pays for the roads. **Who leaks, and whether they accept the patch, is pages 3 to 5.**

→ *So Norway needs the dynamic layer TØI calls for. What is it, and who has built the pieces?*

A dynamic layer beside the national models

The national system has three parts a candidate must know. NTM6 is the national passenger model for trips over 70 km. RTM is the regional one. NGM is the national freight model [toi1819, ngm]. Kristiansund is a regional, sub-70 km case, so RTM is the model this layer sits beside here; NTM6 and NGM set the national and freight context. TØI's own review asks for exactly this. Not to replace these models, but to build dynamic ones next to them [toi1819]. The institute has started, with a MATSim assignment for Oslo and an activity model for Trondheim [tramodsim]. This PhD joins that line. It adds the layer that stress-tests instruments across thousands of futures and reports where each one fails, so the national models keep giving the agencies their appraisal while this layer gives the robustness they were never built for.

THE POSITION ASKS FOR	THIS PROPOSAL ANSWERS WITH
The zero-emission transition, with its effect on finance and equity	Emissions, the eroding toll base, and a distributional floor are three explicit 2050 targets
Behaviour under deep uncertainty	1,000 futures, satisficing targets, uncertainty split by source
DCM joined to ABM	Choice models from real survey data drive agents whose acceptance and fleet carry over year to year
Backcasting, not one forecast	2050 targets fixed on page 3; instrument orders searched backward from them
A practical contribution	Three testable claims, and a named decision on the zero-emission discount and tariff

The gap. Strategic models give end-states. NGM gives freight chains. Neither carries acceptance. Norwegian agent work assigns traffic with MATSim but takes acceptance as given [tramodsim]. Le Pira, Marcucci and Gatta put discrete choice and agent models together to test the acceptability of urban freight policy [lepira, marcucci17], but their agents are stakeholder groups bargaining in rounds, not a population living through the years. Adaptive-pathway methods sequence actions against tipping points [haasnoot], but treat acceptance as an outside scenario. The deep-uncertainty frontier runs ensembles over agent models, most recently on a made-up city with acceptance left out [fuhrer]. The acceptance evidence itself is careful, and mostly static [hybridaccept, odeck08, borjesson16, toi2179].

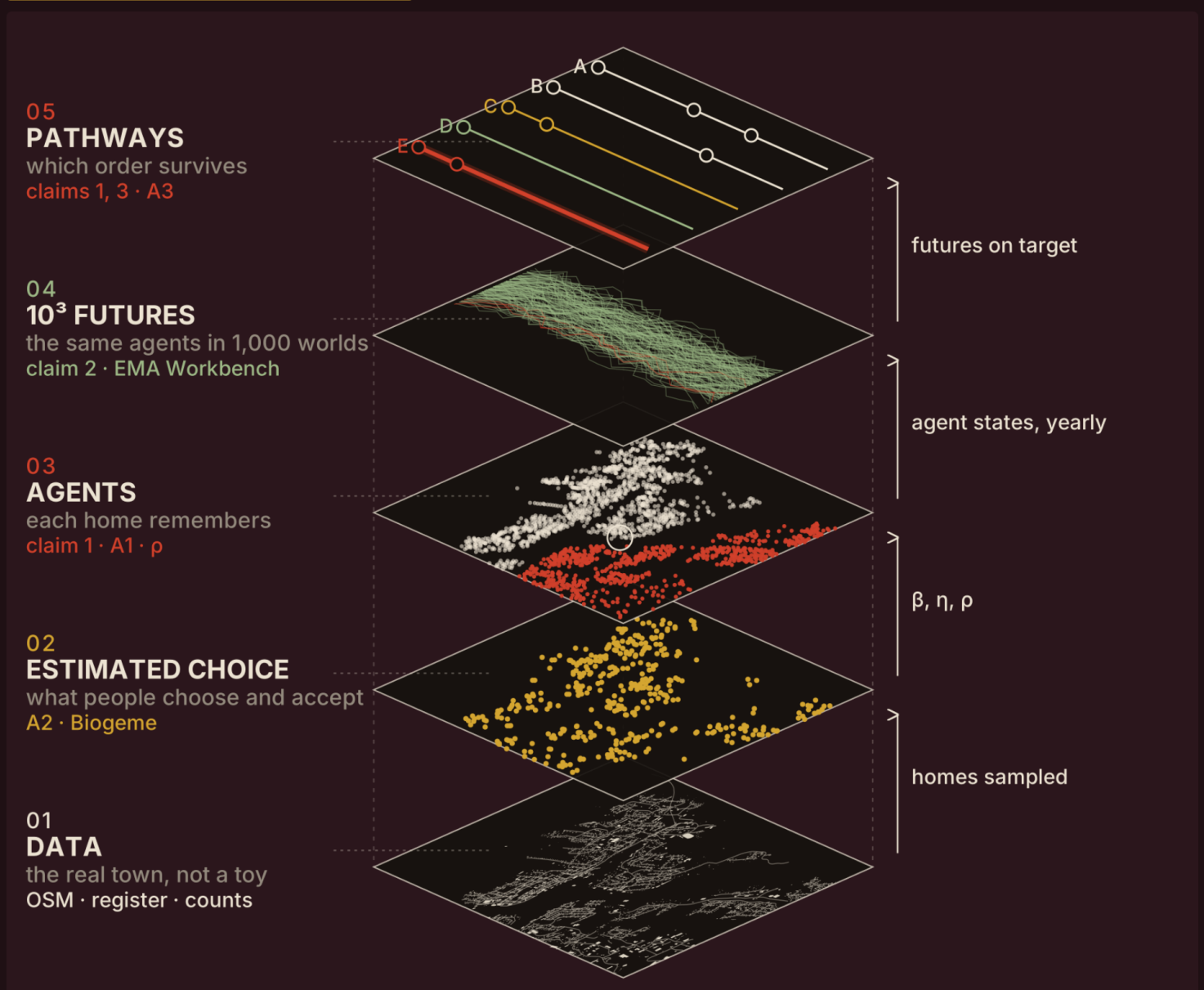
> *So the contribution is one line. No one has estimated acceptance on real data, carried it in agents through calendar time, and let it decide which passenger and freight pathways survive. The prior question is simpler still: is acceptance even measurable? The thesis answers that first. Everything else is what the answer is used for.*

1
Acceptance, estimated and carried
A pathway counts only if estimated acceptance stays above a line every year. The biggest shift it survives is its acceptance margin. A result, not an assumption. (Page 5.)

2
Uncertainty split by source
The 2050 spread is split, year by year, into behaviour, outside futures, and the choice of path. Planners see which uncertainty more data can cut, and which only adaptation can. (Page 5.)

3
Signposts, tested twice, ordered across modes
Warning signals are scored on held-out futures, then against events that happened. Passenger and freight signposts are compared, because the slower fleet decides which fires first. (Pages 4 to 5.)

THE ENGINE, AND WHAT EACH LAYER CARRIES



Read it bottom to top. The real town (OSM, register, counts). Choice estimated from survey data (Biogeme). Agents that remember, carrying acceptance through ρ (the custom agent model). The same agents in a thousand futures (EMA Workbench). The order that survives (the robust search). **Freight rides all five: carrier choice is estimated too, heavy vehicles carry their own fleet and load factors, and freight pathways are searched beside passenger ones.** *An equilibrium flattens every plane into one number for one year.*

→ *Here is that layer, written down.*

What each agent decides and remembers

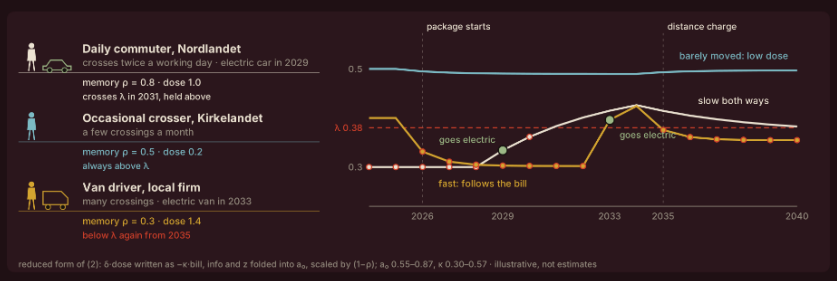
Each year, a household or a firm picks a vehicle, how often to cross, and when. The instruments in force set the cost. Acceptance is estimated first in its plain cross-sectional form (A1), as an integrated choice and latent-variable (ICLV) model on real wave-1 data. This is the hybrid-choice approach used to read acceptance surveys elsewhere [hybridaccept], carried from the bargaining stakeholder groups of Le Pira, Marcucci and Gatta to a whole population of individually estimated agents [lepira].

```
(1)  U(n,j,t) = β·cost(j, I(t)) + θ'·x(n,j) + ε(n,j,t)
(2)  A*(n,t) = ρ·A(n,t−1) + δ·dose(n,t) + η·info(n) + γ'·z(n) + u(n,t)
(M)  I_k(n) = α_k + λ_k·A*(n) + ν_k  (attitudinal indicators load on latent acceptance)
(3)  a pathway is admissible if the share with A* ≥ λ holds every year; its acceptance margin is the largest rise in λ it survives.
```

Acceptance is a hidden variable. We measure it through survey questions: is the charge fair, is it trusted, is it worth its purpose? Equation (M) ties those answers to it, and the model estimates acceptance jointly with people's choices. Tastes vary across the population, so β is random with a normal spread. This is the full ICLV, not a simple attitude score. The dynamic form (2) adds one memory term, ρ, found between the two survey waves with the initial-conditions problem handled [wooldridge]. **H1 tests ρ and H2 tests η, both on page 4.** ρ is the stretch, pursued only if both waves land. The core rests on the cross-sectional model.

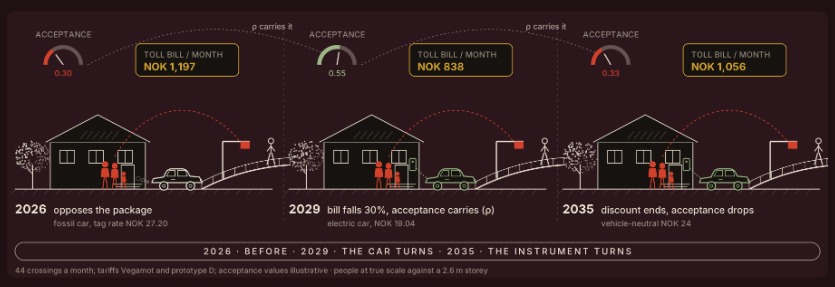
The estimated numbers then drive a custom agent model in Python and NumPy. Each year, agents replace cars by a binary logit, choose modes by the estimated MNL, and hit congestion on a BPR curve solved by successive averages. That is the engine that carries acceptance and fleet forward, year by year. **It already runs, end to end.**

WHAT MEMORY DOES



Three agents to 2050. The commuter crosses the line λ in 2031 and, because ρ is high, stays above it. The occasional crosser never leaves. The van driver, with the biggest dose, falls below λ from 2035. **Memory, not this year's bill, decides who stays accepting. That is the thing an equilibrium cannot hold.** *ILLUSTRATIVE · ρ, dose not estimates*

ONE HOUSEHOLD, THREE MOMENTS



2026: the household opposes the package, on a fossil car. 2029: it buys electric, the bill drops 30 percent, and acceptance carries forward. 2035: the discount ends, and acceptance drops. **The same ρ that holds acceptance up in 2029 is what lets it fall in 2035. Memory cuts both ways.**

What a pathway is. Five instruments: revising the package, the zero-emission rate, a passenger distance charge, a heavy-vehicle distance charge, and earmarking the revenue. A3 is a search, not a catalogue. It picks each instrument's switch-on year and level to put the most futures on all three 2050 targets and to minimise the worst-case regret, while keeping acceptance above the line every year, revenue above debt service, and access above a floor. A many-objective robust algorithm does the search; scenario discovery names where the robust set still breaks [prim, mordm]. Robustness as the thing to optimise follows the stability-metrics line of Shaabani, Hvattum, Hoff and Laporte [shaabani]. A reduced set of four instruments enumerates to 4 + 36 + 216 + 648 = 904 ordered sequences, kept as a check; the full search over years and levels runs to about 10^4 pathways.

TARGET IN 2050	METRIC	ANCHOR
Emissions	Well-to-wheel CO_2e from road traffic in the case area	Climate Change Act: 90 to 95% by 2050 [klimalov]
Finance	Toll revenue at or above debt service; failure if below for three years	Package financing [innst337]
Access (equity)	Generalised cost of the lowest income quintile no higher than 2026	The equity the call names; page 4 shows its spatial side

What would prove it wrong. If ρ is near zero, agents need no memory and claim 1 collapses to a static result. If η is near zero, information does not move acceptance and H2 fails. If the fast engine cannot match the network model, the ensemble shrinks to what the network model can run. Each is reported whichever way it falls. **Tools.** Python throughout: Biogeme for estimation, the custom agent model to carry state, EMA Workbench for the futures and the search, OSMnx and NetworkX for the networks, OR-Tools for routing, NGM for freight. The whole chain already runs. As a check I reproduced a standard mixed logit on the public Swissmetro data in Biogeme: final log-likelihood −5214.9 against the published −5214.8 [evdcm]. The code is open and tested.

→ *Tested where acceptance is live and freight is real.*

04 · CASE AND DESIGN

Kristiansund, where acceptance is live

Why Kristiansund, and where freight enters. Bypakke Kristiansund began collecting on 1 June 2026, at two rv70 stations, Nordsundbrua and Omsundbrua. About 70 percent of the package is toll-financed, over up to 15 years [bypakke, innst337]. A car pays NOK 34, or 27.20 with a tag, or 19.04 if it is zero-emission with a tag [vegamot]. The council approved the basis 26 to 19, the county 33 to 14. Acceptance here is contested, and live. The town sits on the Møre og Romsdal corridor, where the Ferjefri E39 programme and NGM both treat coastal short-sea as the competing mode for freight bound to Oslo and the continent [ngm], and whether that mode is actually cleaner turns on load factors, not the label [hjelle]. Freight is estimated here, not borrowed. A short carrier choice block asks hauliers how they would move, and when, under a distance charge, and the NGM chains carry that answer to the network. The heavy fleet turns over slowly and the sea route needs volume to stay viable, so a pathway that charges road freight too late shuts the modal-shift window and locks the emissions in. **Molde, with no package, is the no-toll control town:** information without experience, set against Kristiansund's information plus experience.

THE TOLL IS A MAP OF WHO CAN AVOID IT



Every building is tied to the side of Nordsundbrua it can reach without the bridge, on the real OSM graph. Red is Nordlandet, which only the bridge connects: 48 percent of junction-pairs lose every road link at the split. Teal is the toll-free Atlanterhavstunnelen, a free way out for some. **Exposure is geography. The equity target on page 3 is income. The thesis reports one crossed with the other, so the floor is about who pays, not just where.** *REAL · OSM graph*

The survey: three blocks, Bayesian D-efficient

Acceptance block (households). Information is randomised in three arms: personal cost, a short purpose note, and a control. So both the content and the framing are tested. **Passenger choice block.** Vehicle at next replacement, crossing frequency and time, at national tariff levels. **Carrier choice block** (firms and hauliers). Road against coastal short-sea, route and timing under a distance charge. One wave, a small targeted sample, anchored to NGM, reached through freight-industry channels. Kristiansund carries the household and carrier design; Molde the acceptance control. Wave 1 and the carrier block in month 6; wave 2 in month 18, same households. The planning sample is 1,428 in Kristiansund and 714 in Molde, set by the smallest effect worth detecting.

Three hypotheses

H1. Acceptance lasts once a charge is permanent ($\rho > 0$ in eq. 2), tested as within-person change between waves against dose. **H2.** Information moves acceptance on its own ($\eta \neq 0$ in eq. 2), tested by the randomised arms and the Molde control, turning the Norwegian link between information and attitude [odeck08] into a causal estimate. **H3.** A distance charge pushes carrier mode choice toward short-sea above a threshold rate.

Ethics and data

The two waves re-link the same households twelve months apart, so identifiers are kept for a year. Lawful basis under GDPR Article 6; consent to re-contact taken at wave 1; the key held apart from the answers, by the supervisor; a written retention-and-deletion schedule; attitudinal data kept to a minimum. The study is notified to Sikt before any fieldwork, with a Data Management Plan filed. Traffic data are used as aggregate station counts only. SSB and RVU microdata are applied for in week 1, under their own agreements.

If the freight survey comes up short

The freight arm then stands on NGM's mode-split elasticities, with the key elasticity swept and reported. The freight signpost rests on a national model instead of a local estimate. It is a floor, not a retreat. The freight result holds either way.

MOLDE · THE SAME METHOD, ON A SECOND REAL NETWORK



Route a van between every pair of junctions on Molde's real road graph. 18 percent of all fastest routes funnel through one street, Øvre veg. Close it and 32 percent of trips slow, 0.9 percent strand. Kristiansund's bridge is the same fragility, 48 percent of junction-pairs cut. Measured on two real towns, not assumed. *REAL · Molde OSM · betweenness*

→ And here is how it finishes in three years.

How it finishes in three years

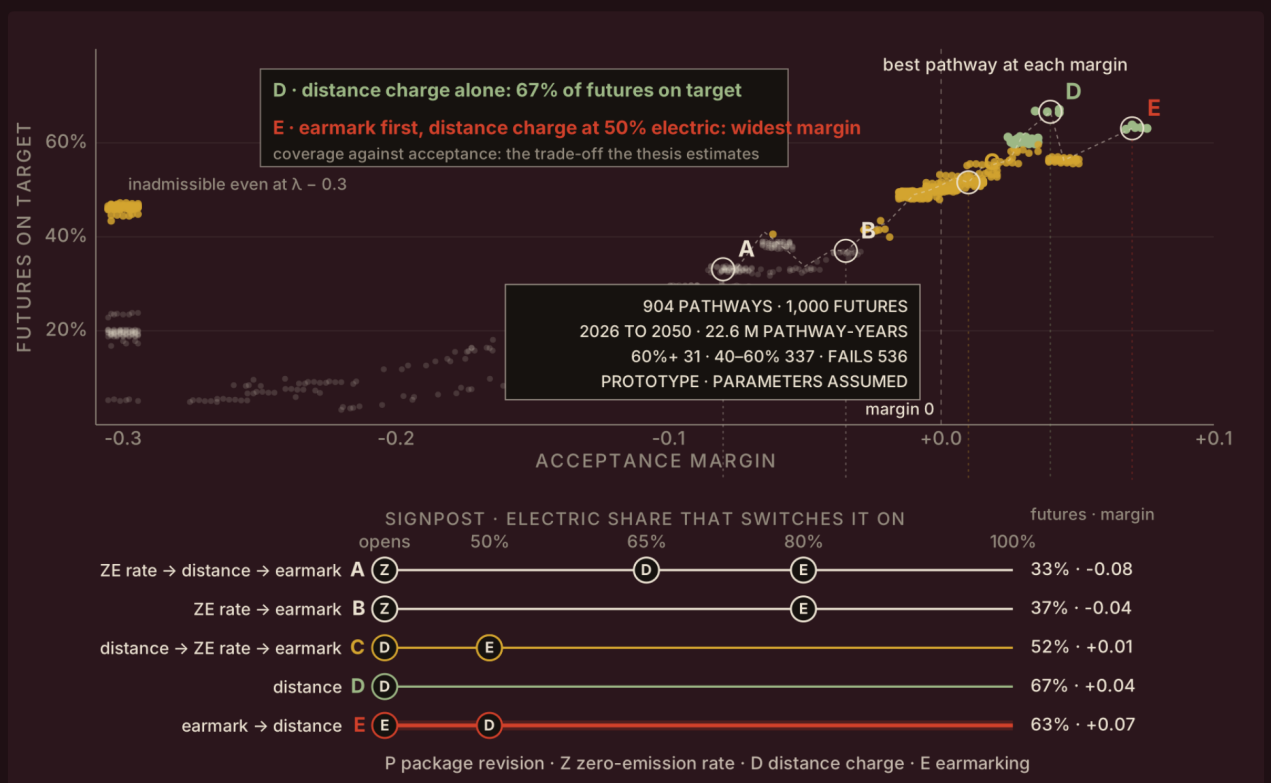
A thesis that survives a failed survey

Level 1	No new data. A1 on national ranges, A3 on NGM elasticities, freight included. A complete thesis on its own.
Level 2	Wave 1 and the carrier block. Adds H2, a cross-sectional A2, and estimated freight.
Level 3	Both waves. Adds H1 and the memory term ρ, the stretch.

Each level is a finished dissertation. The survey only raises the ceiling. The guaranteed floor is my own collected data, under my control. That is how this finishes in 36 months, even if the microdata or a survey wave never arrives.

The search, priced. A3 explores about 10^4 candidate pathways, each scored against a Latin-hypercube sample of futures, and the robust set is re-checked on the full 1,000. Each run advances a zonal state to 2050 in about 25 ms, so the budget is 40 to 150 CPU-hours, parallel, inside a desktop over a weekend. Fidelity is measured on two real networks, Molde and Kristiansund: the fast engine matches static-network car times per zone pair to about 15 percent (r 0.73 to 0.84). A dynamic MATSim cross-check (Molde, 4,000 agents) matches to about 19 percent at free flow and to 6 percent on the static assignment, and the free-flow engine breaks under a tenfold capacity squeeze. That is why robust pathways are re-checked on the network model [evfid].

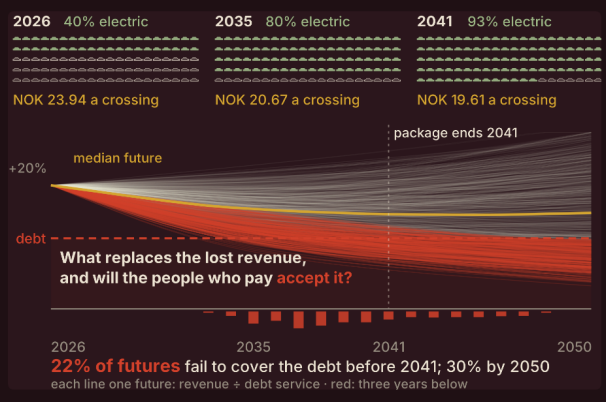
THE ROBUST FRONT, AND THE ORDER EACH PATHWAY SWITCHES ON



Each dot is one of 904 pathways, scored on acceptance margin and share of futures on target. The front is the best trade-off; E is chosen for the widest shift in λ it survives. The rail below is the finding the scatter cannot show. **The same instruments in a different order give a different fate.** *A opens with the zero-emission rate and fails. E earmarks first, then charges distance, and holds. The order is the result.*

PROTOTYPE · parameters assumed until A1 to A3 · 10 seeds · hypervolume-converged

WHAT REPLACES THE LOST REVENUE



Revenue over debt across 1,000 futures. 22 percent fail to cover the debt before 2041, 30 percent by 2050. **This is the page-1 erosion grown up. The minus-30-percent-a-crossing becomes a debt-cover signpost, and the year the median future crosses the debt line is the finance signpost the ring promised.** PROTOTYPE

CLAIM 2 · UNCERTAINTY BY SOURCE

Where the 2050 spread comes from, by year. Planners see which uncertainty more data can cut (behaviour) and which only adaptation can (outside futures).



Sobol variance shares, prototype. Behaviour dominates early, so more survey data helps. Outside futures dominate late, so only adaptation does.

PROTOTYPE · illustrative

ARTICLE	SUBMIT	DATA
A1 Choice estimated	M12	Wave 1 + carrier block
A2 Acceptance that lasts	M24	Both waves
A3 Robust pathways	M30	Ensemble; dated events

The computational stack, one tool per article

ESTIMATE	CARRY	EXPLORE	NETWORK	FREIGHT	COMPOSE
Biogeme mixed logit, ICLV Acceptance and carrier mode choice, from real survey data. A1, A2	**Agent model** Python, NumPy Fleet logit, estimated MNL, BPR congestion. Acceptance and fleet carried year to year. A2, A3	**EMA Workbench** ε-NSGA-II, PRIM 1,000 futures, robust search, scenario discovery. A3	**OSMnx, NetworkX** OR-Tools, MATSim Real road graphs, routing checks, dynamic cross-check. A1, A3	**NGM** national freight model Commodity chains, mode split, load factors, empty running. A3	**Rhino, Grasshopper** Ladybug, Blender Every figure computed from the data, in one parametric pipeline. all plates

Calendar: 36 months, full research

Week 1: the SSB and RVU applications and the Sikt notice go in, off the critical path. Months 1 to 6: coursework, including philosophy of science and discrete-choice courses, and a pilot with Norwegian-speaking co-authors. Month 6: wave 1 and the carrier block. Month 12: A1. Month 18: wave 2. Month 20: midway evaluation. Month 24: A2. Month 30: A3, with six months left for the summary chapter and the defense. Articles are pipelined, not serial: A1 is under review while A2 runs.

Risk and response

RISK	RESPONSE
Low response or attrition	Panel top-up; sample sized for it; Level 2 holds
No microdata by M6	Published ranges and NGM; the spine is my own survey
Carrier survey short	Stand on NGM elasticities, sweep the key one
Language	Instrument in English, translated and piloted with Norwegian co-authors

Supervision and fit

The supervision this work needs is resident in the department. Robustness and stability metrics: the group behind [shaabani], Hvattum and Hoff. Discrete choice: the tradition of [lepira], Marcucci and Gatta, the one skill I set out to deepen. Freight and toll economics: the ground of [odeck08], Odeck and Bråthen, of Hjelle on freight and short sea [hjelle], and the Logistics department itself. The three works this proposal leans on hardest were each written inside Molde's own groups. All the modelling and evidence so far is my own.

Why here, and the decision it yields

The position sits in the Department of Logistics and asks for behaviour under the zero-emission transition, with its effect on finance and equity. That is this proposal's spine. The thesis ends on a concrete recommendation: a revised zero-emission discount and tariff for Bypakke Kristiansund, not just a method. **Receipts.** Every number traces to a tagged commit, with 89 automated tests, and every figure is drawn by the same code. The code is open; anonymised data is deposited FAIR once Sikt allows; AI-tool use is disclosed. **Dissemination.** Three journal articles, a Norwegian-language piece in *Samferdsel*, and the annual NORS conference, where I have presented before.

→ The finance signpost that warns a path is failing is the same erosion page 1 showed already under way. The ring closes.

References & receipts

All cited sources verified against primary sources as of 2026-10-17. Evidence entries are runnable artefacts in the repository, traced to tagged commits.

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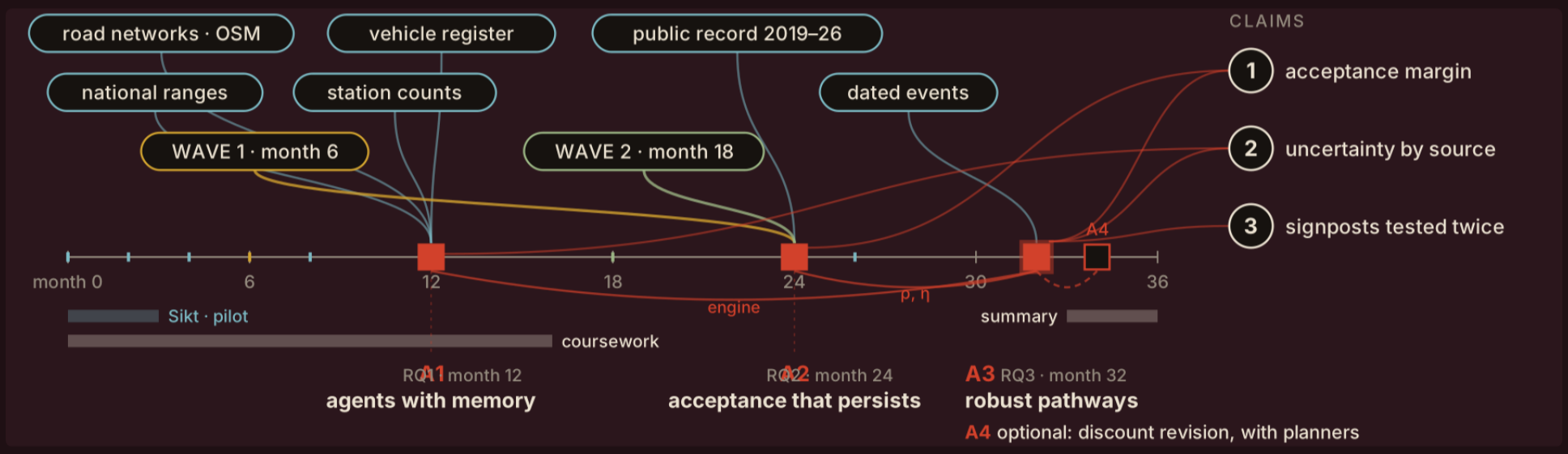
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EVIDENCE LEDGER: RECEIPTS YOU CAN RUN

[evdcm]	**Swissmetro mixed-logit reproduction.** Biogeme 3.3.5, N=6,768, 1,000 Halton draws. Final LL −5214.9 vs published −5214.8. evidence/dcm-swissmetro
[evfid]	**Zonal-vs-network fidelity benchmark.** Molde + Kristiansund OSM graphs. About 15% MAPE (r 0.73 to 0.84); MATSim dynamic cross-check 19% free-flow, breaks under 10x congestion. evidence/fidelity-benchmark
[tests]	**89 automated tests.** Every figure generated from the same code; FAIR data deposit once Sikt allows. pytest, tagged at submission

THE THESIS AS A SYSTEM: DATA TO ARTICLES TO CLAIMS



Each data source feeds a wave; each wave feeds an article; each article delivers a claim. **Cut the survey wires and the teal spine, my own data and national ranges, still reaches A1 and A3. The thesis completes with no outside dependency.**